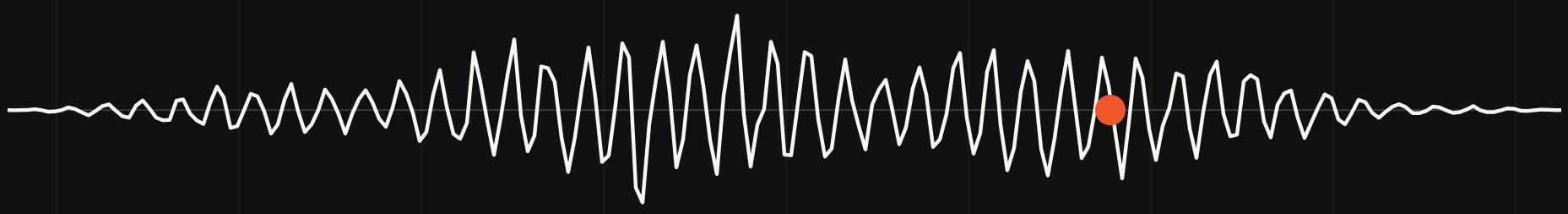
§09 · THE EXAMINER ASKS



"You're turning the engine and water comes out of an indicator cock. What do you do?"

SWIPE →

§09 — FIRST — FIG. 09.2

09.2

Stop. Don't fire that unit.

Water doesn't compress. Fire a unit with water on top of the piston and you risk hydraulic lock and serious mechanical damage, typically to the connecting rod or the cylinder cover studs.

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§09 — THEN DIAGNOSE — FIG. 09.3

09.3

Condensation, or a leak?

LIKELY CONDENSATION	LIKELY A LEAK
A small amount of clear water, on a cold engine, after a long shutdown.	A steady flow, or water with oil or fuel in it.

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§09 — POSSIBLE SOURCES — FIG. 09.4

09.4

Where could it be coming from?

01 Cracked cylinder head or liner

02 Failed liner O-ring or cooling jacket seal

03 Leaking exhaust valve cage seal

04 On some designs, fuel valve cooling

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§09 — WHAT YOU'D DO — FIG. 09.5

In order.

01 Inform the ECR or duty engineer.

02 Check the jacket water expansion tank level, and how fast it's dropping.

03 Boroscope the liner and exhaust valve.

04 Compare unit performance.

05 Rectify before the unit goes back into service.

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§09 — WATCH OUT FOR !

The answer that fails.

- ! Firing the unit to see what happens.
- ! Forgetting the expansion tank, the quickest way to localise a cooling leak.

EXPECT THESE FOLLOW-UPS

- Q1 How do you tell condensation from a leak?
- Q2 What would you check on the expansion tank?
- Q3 What's the risk of firing the unit?

§09 — NEXT STEP — WAITLIST OPEN

Could you answer that out loud?

Find out before the exam. Join the waitlist.

JOIN THE WAITLIST →

ORALROOM.APP/WAITLIST

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